National University of Computer and Emerging Sciences, Lahore Campus

Course:	**Object Oriented Programming**	**Course Code:**	**CS-1004**
Program:	**BS (CS)**	**Semester:**	**Spring 2026**
Deadline:	**1st March 2026**	**Total Marks:**	**25*5 = 125**
Section:	**BCS-2C,2D,2E**	**CLO**	**1**
	Assignment-1		

Instruction/Notes:

Please read the following instructions carefully before submitting the assignment.

1. Save your file with your Roll Number.
2. There should be only one cpp file.
3. It should be clear that your assignment will not get any credit if:
   The assignment is submitted after the due date.
   Assignment is copied (partial or full) from any source (websites, forums, students, etc.)

Important Note:

1. There shouldn't be any memory leakage in your program.
2. Make separate functions for input and output. Your main should be a sequence of function calls only.
3. You are not allowed to use global variables and goto instruction.
4. All allocations of 1D/2D pointers should be dynamic.
5. Delete the array when it is no longer needed.
6. All the data will be given by user.
7. Pass the pointers to functions instead of subscript notation [].
8. Make proper functions to solve the problems.
9. Debug your code to find errors/bugs.
10. Be confident enough that you can do it.

Task 1: [Tokenization]

Prompt the user to enter the data and get the input using character array (input should be at-lease 4 to 5 sentences long where each sentence ended with dot. You can declare a 1D character array (dynamic or static up to you, preferably dynamic) large enough to store 2000 to 3000 characters). It is better to read the data from a text file to save the input time. Your task is to Tokenize the words (here blank space should be treated as a token) entered by user and store these words in another 2D dynamic character array "data dictionary". Make sure there should not be any duplication in data dictionary. Number of rows in data dictionary should be equal to the unique words entered by the user and number of columns in each row should be one greater than the word length to store null terminator.

Example: 1 **Input:** I love Pakistan. **Output:** I Love Pakistan	**Example#2 Input:** This assignment is a piece of cake. **Output:** This assignment is a piece of cake

Task 2: [Synonyms]

Now you need to create a 3D character array "synonyms" equal to the size of unique words in data dictionary and initialize each index of "synonyms" with null pointer. Your program should iterate for each word of "data dictionary" and ask the user whether he/she wants to store synonyms of that word. If user is interested to store the synonym then ask for the count of synonyms i.e., how many synonyms he/she wants to store against that word. Get the count of synonyms from the user and allocate the required memory at that index of "3D synonyms array". Now iterate to get the input for the synonyms. To get the input for synonyms you can use a "temp 1D static character array". Once you get the synonym from user in a "temp" character array you need to allocate the memory at that particular index of "3D synonym array" equal to the size (strlen+1) of "temp" character array then copy the data character by character and then prompt the user to enter the next synonym if any. You need to repeat this task for each word in "data dictionary".

Example:1

Input Word: love
Synonym: affection

Example:2

Input Word: assignment
Synonym: assessment

Task 3: [Words in data dictionary]

Now create another 1D dynamic character array "useCase1" sufficiently large enough to store two to three thousand characters. Prompt the user to enter the data (at-least 3-4 sentences), your program should search each word of this "useCase1" in "data dictionary" and keep the count of words that are matched. You also need to print those words on console that are matched.

Bonus: Include the unmatched words of "useCase1" in "data dictionary".

Example:1

Suppose the string "I love Pakistan" is stored in data dictionary.
Input: They love Pakistan.

Output:
There are 2 words found in data dictionary. The words are as follows:
love
Pakistan

Example:2

Suppose "This assignment is piece of cake" is stored in data dictionary.
Input: This assignment is due by tomorrow.

Output:
There are 3 words found in data dictionary. The words are as follows:
This
assignment
is

Task 4: [Synonym in our stored synonym]

Now you need to replace the words in "useCase1" with their synonyms. First you need to check whether the words in "useCase1" present in "data dictionary". If it is present then you need to check whether there exists any synonym of that word. If "yes" then replace the word of "useCase1" with its synonym.

Example: 1

Input: This is my assignment

Words	**Synonyms**
assignment	assessment

Output: This is my assessment.

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	Course: **Program:** **Due Date:** **Sections:** **Exam:**	**Object Oriented Programming** **BS (CS, DS)**	**Course Code:** **Semester:** **Total Marks:**	

Instruction/Notes:

Instructions: Plagiarism in any form will not be tolerated.

Q#1: You are given a **static 2D character array** of tweets. Each tweet is stored in a row of the array, and each row can have a different length of characters. The number of rows in the array is **N** (the total number of tweets), and the number of columns in each row is **M** (the maximum length of a tweet). Your task is to extract the **distinct hashtags** from the tweets and store them in a new **dynamic 2D character array**. Each hashtag should be stored in a separate row of the array. A **hashtag** is defined as a word that begins with the symbol # and is followed by alphanumeric characters (letters or digits) or underscores (_).

Constraints:

1. **You can not use any built-in function in this homework. Main function should only be used for declaration, initialization and function calling.**
2. There is a possibility that the same hashtag may appear in multiple tweets. However, you must ensure that **only distinct hashtags** are stored.
3. The dynamic 2D char array should have exactly **one row for each distinct hashtag**, and the number of columns in each row should be equal to **the length of the hashtag + 1** to accommodate the null terminator (\0).
4. The input array of tweets (2D static array) is already provided. You can hard code this sample input (or read the data from file) to focus on your actual logic building rather wasting the time input. You are required to:
    - Extract all the hashtags from each tweet.
    - Filter out duplicates, ensuring only distinct hashtags are kept.
    - Store the distinct hashtags in a **dynamic 2D character array**.
    - Ensure that memory is allocated only for the exact number of distinct hashtags, and each row in the array has enough space to store a hashtag including the null terminator.

Hint! You need to resize the hashtags array each time when a distinct hashtag is found. Create a temp 2D char array, copy the previous hashtags, store the new hashtag and relocate hashtags pointer to this new memory. Don' forget to deallocate the memory. (**10 minutes class discussion (10/11 Feb)**).

Sample Output:

Tweets[N][100]; N is a const int (20)

I love programming! #coding #developer #tech
Just read a great book about AI! #reading #AI #books
The weather today is fantastic! #sunny #beautiful
Had an amazing workout session! #fitness #health #motivation
Finally, coffee time! #coffee #morning #energy
Learning C programming! #coding #learning
Can't wait for the weekend! #weekendvibes #excited

Enjoying a relaxing day at the beach! #vacation #beach #relax
New project starting today! #work #project #goals
Running in the park! #fitness #nature
Loving this tech conference! #tech #developer #AI
Weekend vibes and relaxation time! #weekendvibes #vacation #relax
Fitness goals achieved today! #fitness #health #motivation
Finally got my first coffee today! #coffee #morning #energy
Building a new app with C! #coding #developer
Taking a walk at the beach. #beach #vacation #nature
Just finished an AI course! #AI #learning #tech
Work is so rewarding today! #work #project #goals
A morning walk to clear my mind! #morning #fitness #nature
Amazing sunny day! #sunny #beautiful #weekendvibes

Hashtags (dynamic 2D null terminated character array). You need to allocate exact amount of memory like there are 24 distinct hashtags in this sample input so there are exactly 24 rows of hashtags, and the number of columns in each row are exactly 1 greater than the hashtag length(extra cell to store null terminator).

c	**o**	**d**	**i**	**n**	**g**	**\0**						
d	**e**	**v**	**e**	**l**	**o**	**P**	**e**	**r**	**\0**			
t	**e**	**c**	**h**	**\0**								
r	**e**	**a**	**d**	**i**	**n**	**G**	**\0**					
A	**I**	**\0**										
b	**o**	**o**	**k**	**s**	**\0**							
s	**u**	**n**	**n**	**y**	**\0**							
b	**e**	**a**	**u**	**t**	**i**	**F**	**u**	**l**	**\0**			
f	**i**	**t**	**n**	**e**	**s**	**S**	**\0**					
h	**e**	**a**	**l**	**t**	**h**	**\0**						
m	**o**	**t**	**i**	**v**	**a**	**T**	**i**	**o**	**n**			
c	**o**	**f**	**f**	**e**	**e**	**\0**						
m	**o**	**r**	**n**	**i**	**n**	**g**	**\0**					
e	**n**	**e**	**r**	**g**	**y**	**\0**						
l	**e**	**a**	**r**	**n**	**i**	**n**	**g**	**\0**				
w	**e**	**e**	**k**	**e**	**n**	**d**	**v**	**i**	**b**	**e**	**S**	**\0**
e	**x**	**c**	**i**	**t**	**e**	**d**	**\0**					
v	**a**	**c**	**a**	**t**	**i**	**o**	**n**	**\0**				
b	**e**	**a**	**c**	**h**	**\0**							
r	**e**	**l**	**a**	**x**	**\0**							
w	**o**	**r**	**k**	**\0**								
p	**r**	**o**	**j**	**e**	**c**	**t**	**\0**					
g	**o**	**a**	**l**	**s**	**\0**							
n	**a**	**t**	**u**	**r**	**e**	**\0**						